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AWS IAM Misconfiguration and Least Privilege Project Report

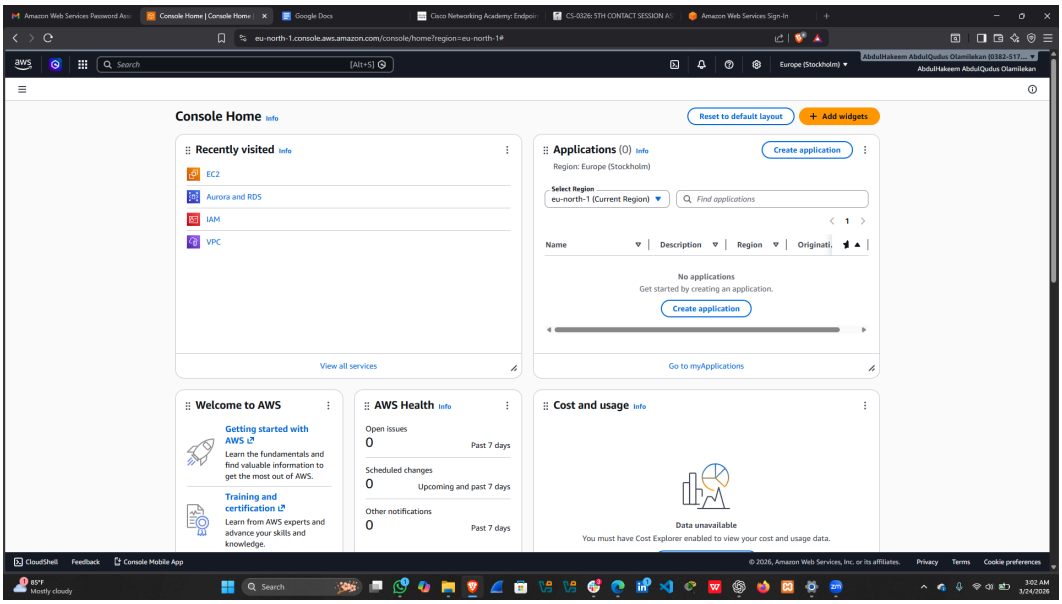
Introduction

This project demonstrates the identification, exploitation, and remediation of an AWS Identity and Access Management (IAM) misconfiguration. It focuses on excessive permissions assigned to an IAM user and how applying the Principle of Least Privilege (PoLP) enhances cloud security.

Objectives

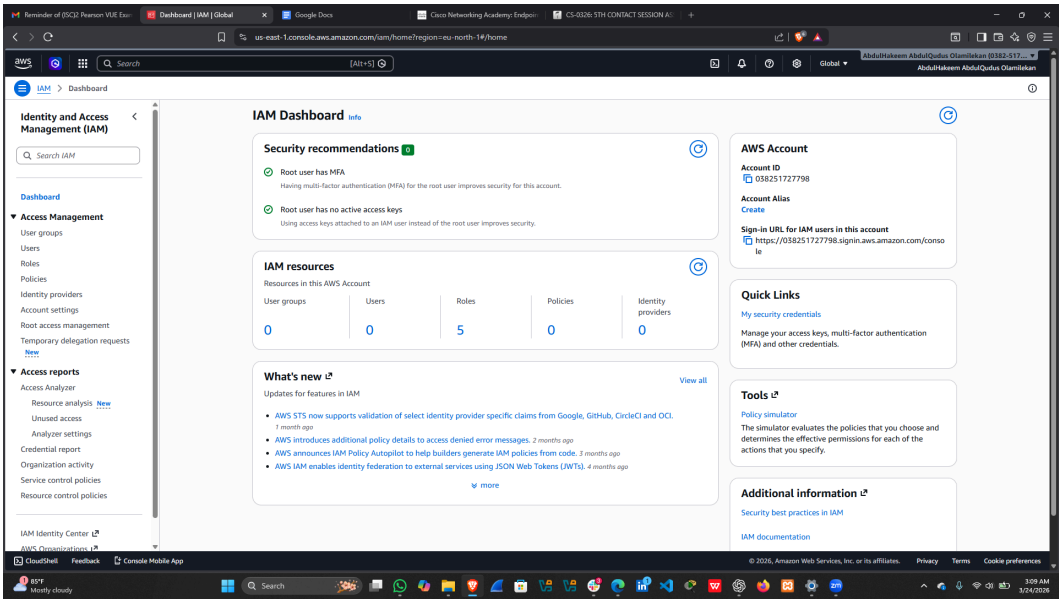
- Create an IAM user with excessive permissions
- Attempt privilege escalation using AWS CLI
- Identify security risks
- Remediate using least privilege
- Validate restrictions and functionality

Environmental Setup



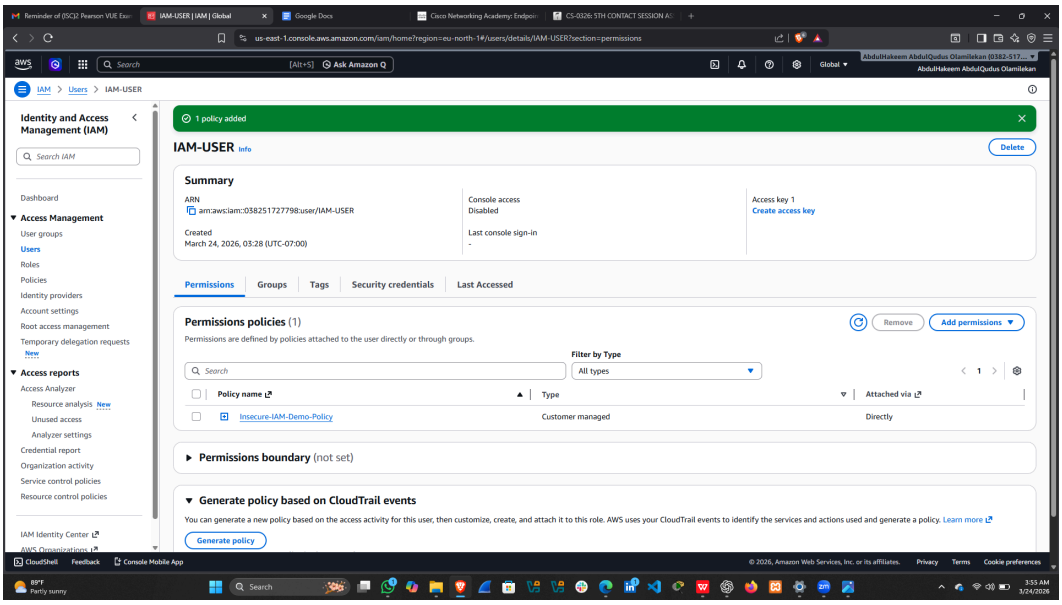
This screenshot shows the stage: Environmental Setup. It demonstrates the relevant configuration or validation step in the IAM security workflow.

IAM Dashboard Overview



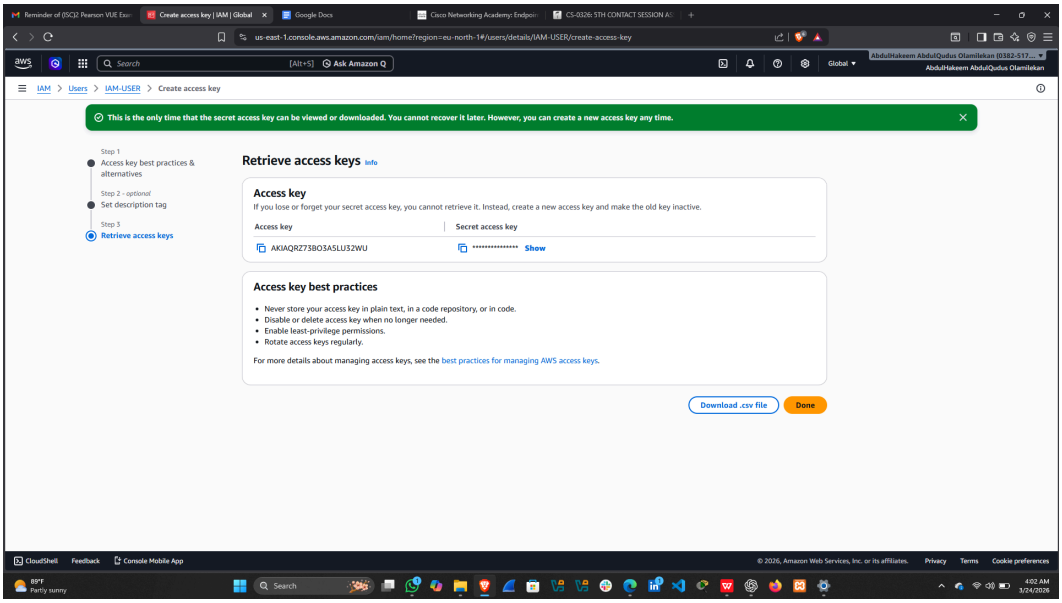
This screenshot shows the stage: IAM Dashboard Overview. It demonstrates the relevant configuration or validation step in the IAM security workflow.

Policy Attached (Insecure Configuration)



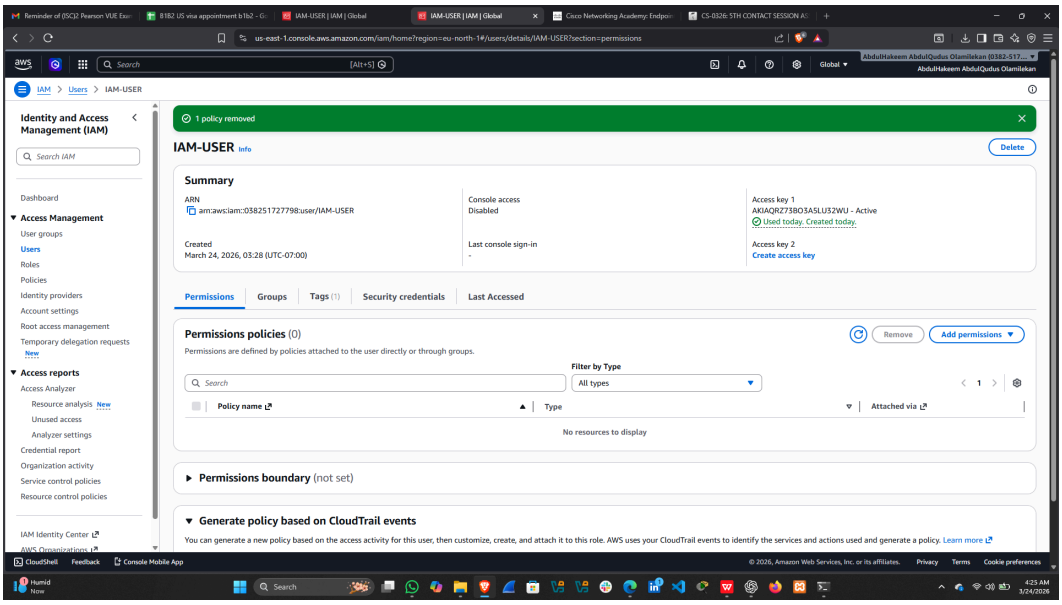
This screenshot shows the stage: Policy Attached (Insecure Configuration). It demonstrates the relevant configuration or validation step in the IAM security workflow.

Access Key Creation



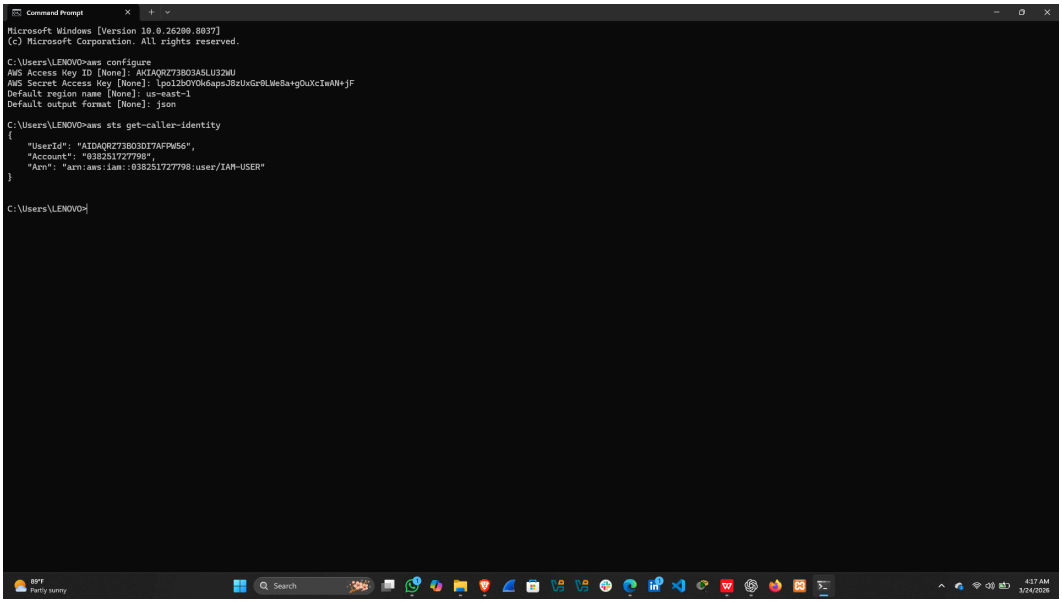
This screenshot shows the stage: Access Key Creation. It demonstrates the relevant configuration or validation step in the IAM security workflow.

Policy Removal (Remediation Step)



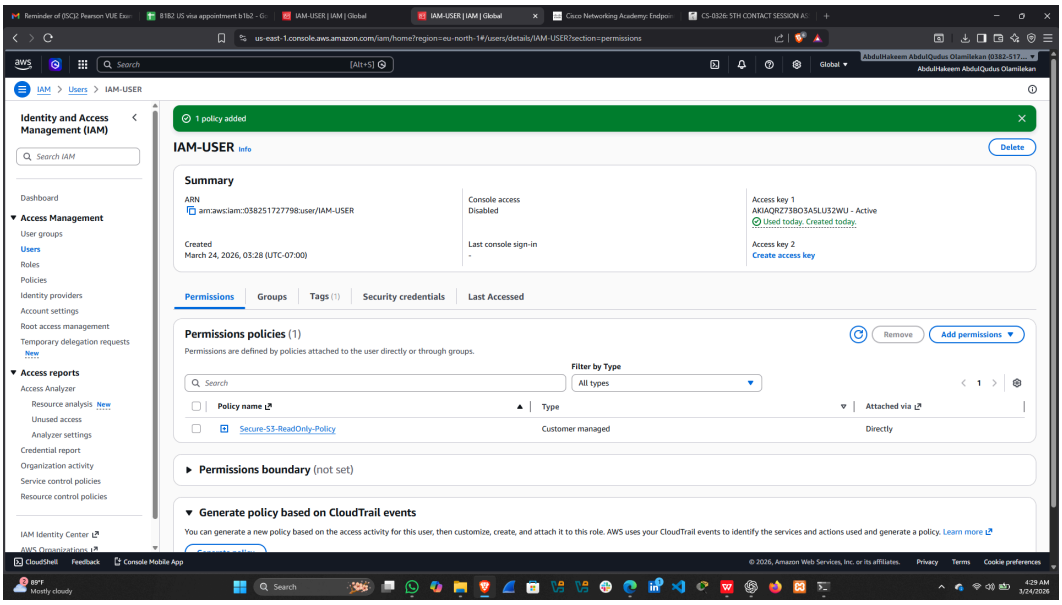
This screenshot shows the stage: Policy Removal (Remediation Step). It demonstrates the relevant configuration or validation step in the IAM security workflow.

CLI Identity Verification



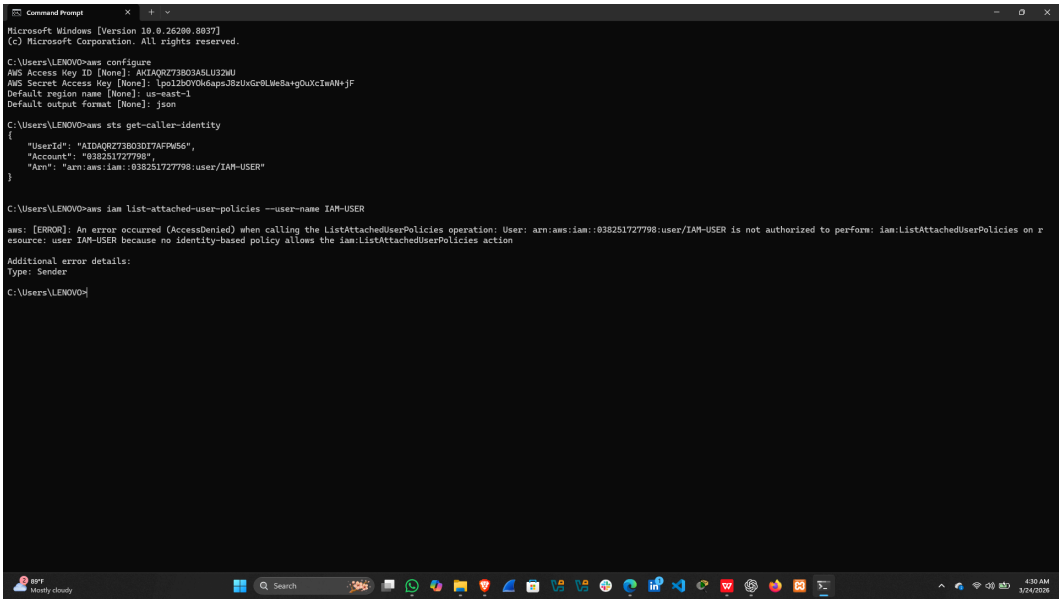
This screenshot shows the stage: CLI Identity Verification. It demonstrates the relevant configuration or validation step in the IAM security workflow.

Secure Policy Attached



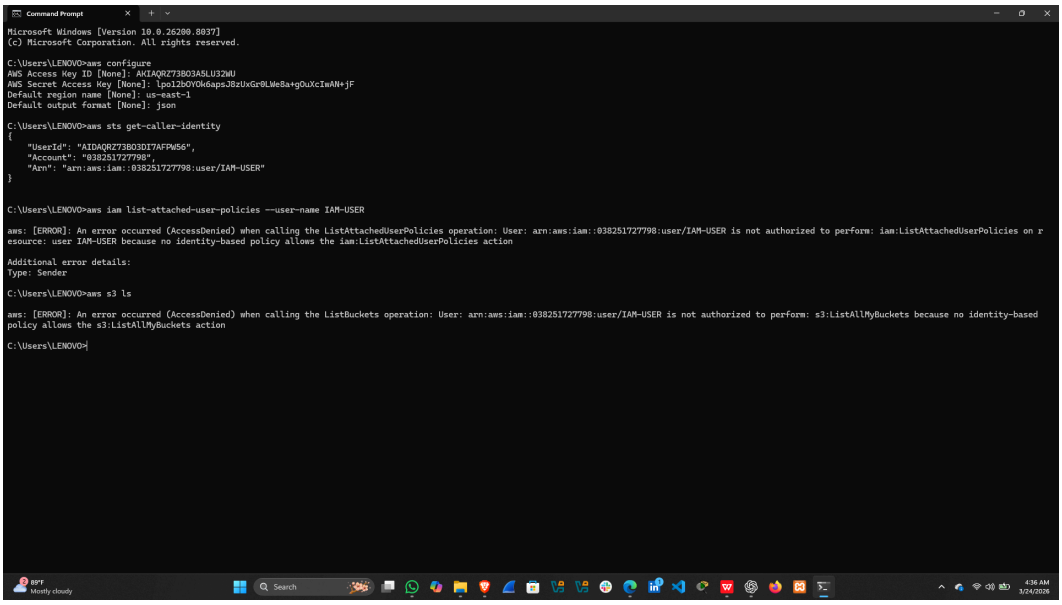
This screenshot shows the stage: Secure Policy Attached. It demonstrates the relevant configuration or validation step in the IAM security workflow.

CLI Access Denied (Privilege Restriction)



This screenshot shows the stage: CLI Access Denied (Privilege Restriction). It demonstrates the relevant configuration or validation step in the IAM security workflow.

CLI S3 Access Test (Initial Failure)



This screenshot shows the stage: CLI S3 Access Test (Initial Failure). It demonstrates the relevant configuration or validation step in the IAM security workflow.

Conclusion

The project successfully demonstrated how misconfigured IAM permissions can lead to security risks. By applying least privilege, access was restricted and validated using AWS CLI, ensuring a secure configuration.